

Weighbridge & Site Access Control Automation System

Complete Software Engineering Diagrams, Data Models, State Lifecycles, and Multi-Tier Architectural Blueprint.

1. Context & Use Case Architecture

Comprehensive actor modeling across Transporters, Weighbridge Operators, Security Officers, Platform Administrators, Drivers, and Autonomous Edge Daemons.

2. Relational Data Model (ERD)

Complete multi-tenant entity-relationship schema encompassing 25+ database models, cryptographic audit chain tables, and edge SQLite schemas.

3. Behavioral & Sequence Lifecycles

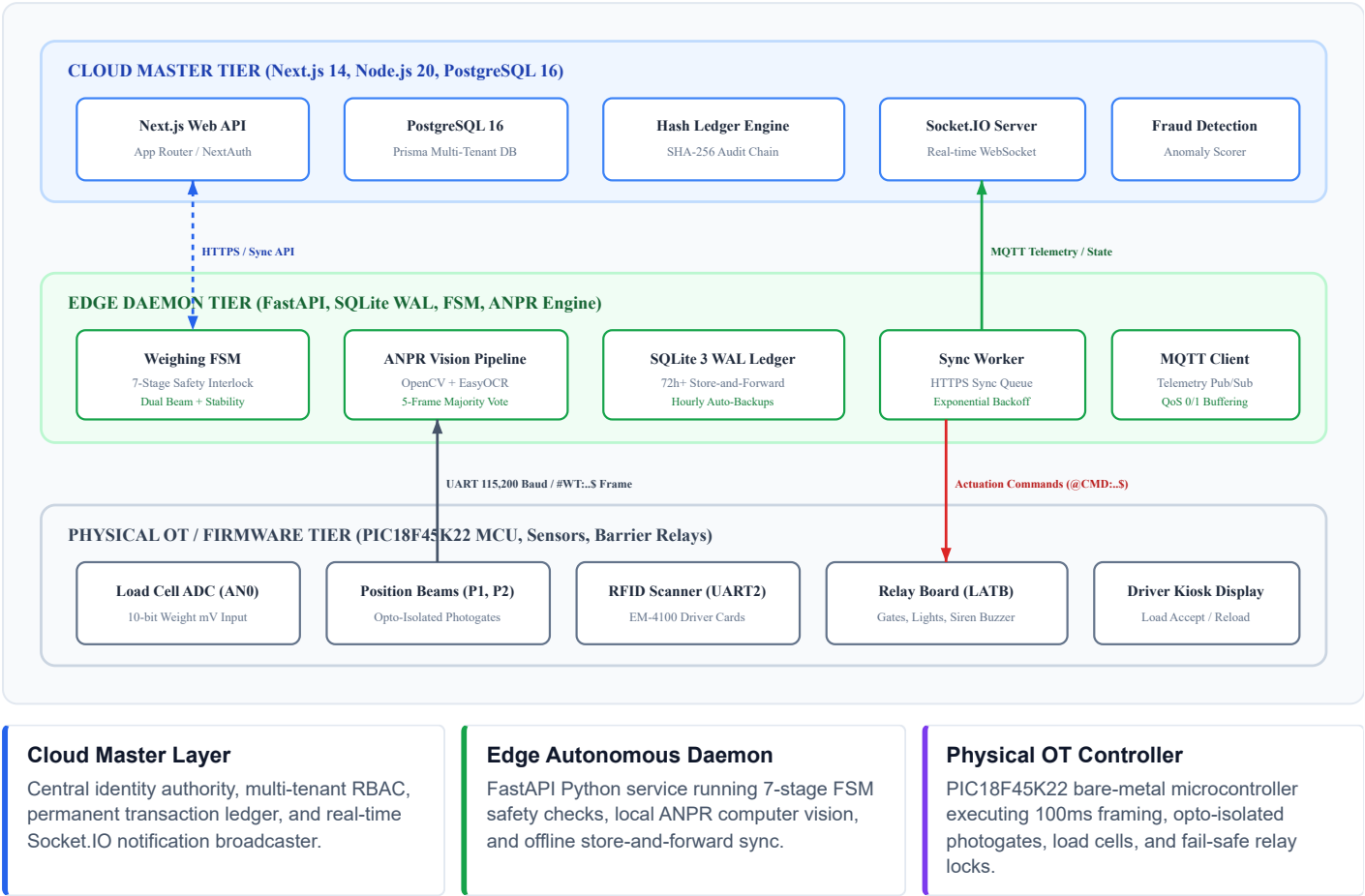
End-to-end automated weighment transactions, offline store-and-forward sync, anti-fraud anomaly scoring, and overload interlock escalations.

4. State Machine & Hardware Protocols

Deterministic 7-stage edge safety state machine, dual optical beam interlocks, UART telemetry framing, and fail-safe relay actuation.

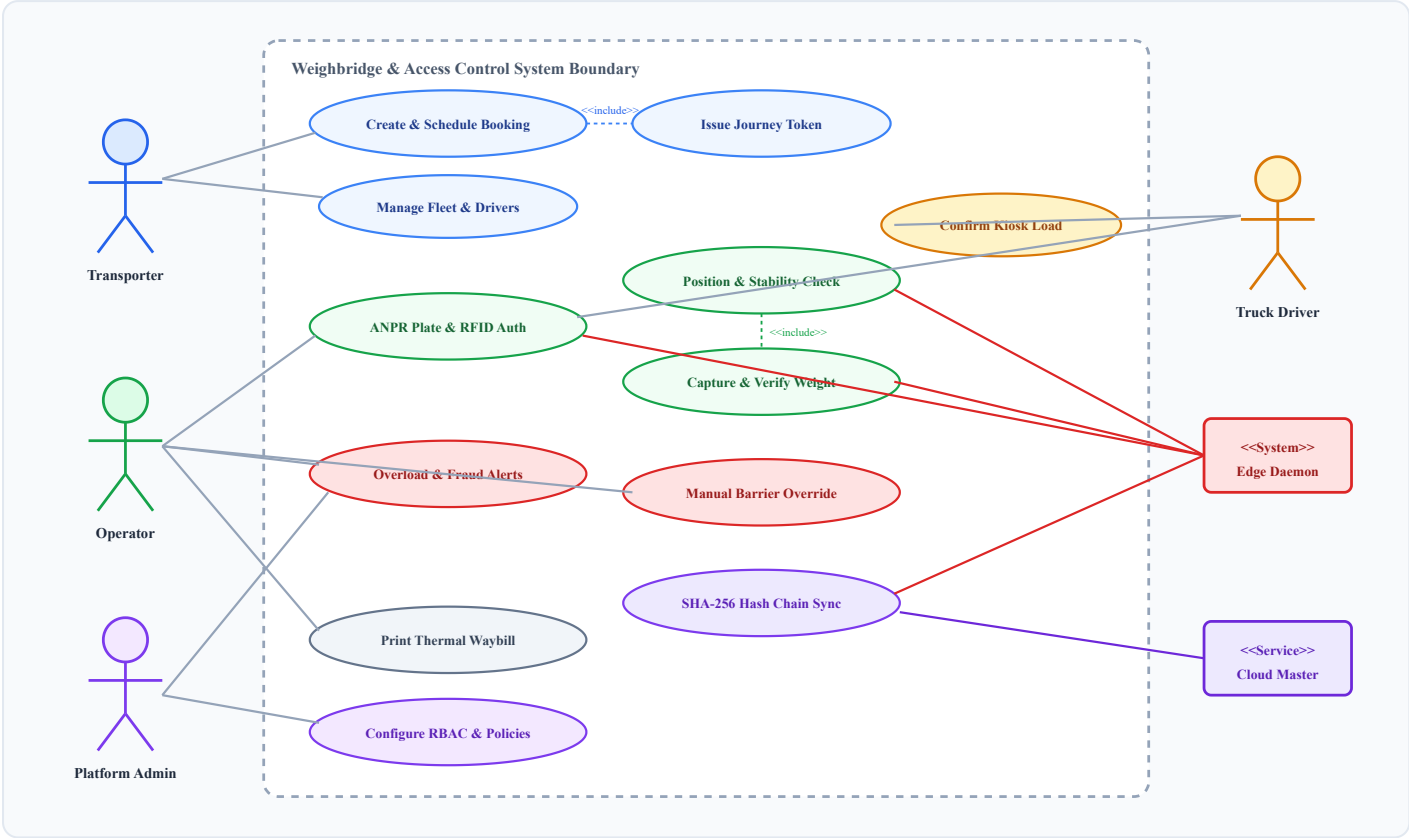
High-Level System Context & Component Architecture

4-Tier physical-to-cloud topology connecting field hardware, edge controllers, real-time message brokers, and enterprise cloud APIs.



Complete UML Use Case Diagram

Actors, core business functions, automated lane interactions, and cryptographic reconciliation workflows.



Key Operational Actors

Transporter: Books truck journey tokens. **Operator:** Supervises live scale deck & resolves alerts. **Driver:** Verifies RFID & accepts load at kiosk.

Autonomous System Actors

Edge Daemon: Interlocks optical beams, executes ANPR OCR consensus, and drives relay outputs. **Cloud Master:** Verifies cryptographic hash chains and recalculates overloads.

Organisation

- PK id: UUID
- name: String
- code: String (UQ)
- type: OrgType
- status: ClientStatus
- taxNumber: String
- isActive: Boolean

User

- PK id: UUID
- FK organisationId
- email: String (UQ)
- passwordHash: String
- role: UserRole
- platformRole: String
- status: UserStatus

Role / Permission

- PK id: UUID
- name: String
- isBuiltIn: Boolean
- M:N RolePermission
- key: String (UQ)
- category: String

Booking

- PK id: UUID
- FK siteId, vehicleId
- FK driverId, trailerId
- reference: String
- journeyToken: String
- commodity: String
- targetTonnageKg: Int
- windowStart, End
- status: BookingStatus

Incident & Fraud

- PK id: UUID
- FK siteId, transactionId
- type: IncidentType
- severity: Severity
- anomalyScore: Float
- evidenceUrls: Text[]
- status: IncidentStatus
- resolvedAt, By

Site & SiteConfig

- PK id: UUID
- FK organisationId
- code: String (UQ)
- topology: Topology
- maxCapacityKg: Int
- stabilityThreshold: 20kg
- overloadTolerance: 5%
- holdSeconds: 2s

Vehicle & Trailer

- PK id: UUID
- FK organisationId
- plate: String
- plateNormalized (UQ)
- tareWeightKg: Int
- legalMaxGvwKg: Int
- insuranceExpiry: Date
- anomalyScore: Decimal

Driver

- PK id: UUID
- FK organisationId
- firstName, lastName
- rfidTag: String (UQ)
- licenceNumber: String
- licenceExpiry: Date
- blacklistStatus: Bool

WeighbridgeTransaction

- PK id: UUID
- FK bookingId, siteId
- FK vehicleId, driverId
- grossWeightKg: Int
- tareWeightKg: Int
- netWeightKg: Int
- waybillNumber (UQ)
- previousHash: SHA256
- integrityHash (UQ)
- confirmationHash
- status, syncStatus

Hardware & Status

- PK id: UUID
- FK siteId, laneId
- deviceType: DeviceType
- health: HealthStatus
- sensorReadings: JSON
- connectivity: JSON
- calibrationExpiry
- lastSeen: DateTime

Edge SQLite Local Ledger & Store-and-Forward Tables (apps/site-daemon/database.py)

- cached_bookings**
 - PK id (UUID)
 - plate_normalized
 - window_start, end
 - payload_json (TEXT)
 - cached_at (ISO)
- transactions (Local Ledger)**
 - PK edge_transaction_id
 - gross, tare, net_kg
 - waybill_number (UQ)
 - previous_hash, integrity_hash
 - sync_status: pending/synced
- sync_queue (Idempotent)**
 - PK id (AUTOINCREMENT)
 - operation: reconcile/upload
 - idempotency_key (UQ)
 - payload_json, retry_count
 - next_retry_at (Backoff)
- metadata & sessions**
 - key: 'last_hash' (Genesis)
 - key: 'waybill_sequence'
 - incomplete_sessions
 - context_json, state
 - PRAGMA journal_mode=WAL

All vehicles, bookings, orders, and roles partition by organisationId with strict client role policies.

WeighbridgeTransaction enforces an unbroken cryptographic chain linking previousHash to current hash.

SQLite sync_queue enforces unique idempotency keys, eliminating duplicate reconciliation on network recovery.

```

graph TD
    subgraph Edge_daemon [Edge daemon]
        subgraph State_Machine [State Machine]
            WeighingState
            AnprEngine
            TelemetryParser_Protocol[TelemetryParser & Protocol]
        end
        subgraph Database [Database]
            EdgeDatabase
        end
        subgraph Services [Services]
            Reconciliation
            FraudDetection
            Access_RBAC[Access & RBAC]
        end
    end

```

PACKAGE apps.site_daemon (Python / AsyncIO)

WeighingStateMachine

- + state: WeighingState
- + booking: ActiveBooking
- + weight_window: deque[tuple]
- + pending_gross_weight: int
- + process(frame: TelemetryFrame)
- + authorise(booking, anpr)
- + is_stable(now: datetime): bool
- + _process_captured_weight(gross)
- + accept_load(): EdgeTransaction
- + request_reload(): void
- + _driver_mismatch(rfid)
- + reset(): void

AnprEngine

- _reader: easyocr.Reader
- + preprocess(image): (thresh, edges)
- + plate_candidates(image): list
- + read_frame(image): (text, conf)
- + recognise(source): AnprResult

TelemetryParser & Protocol (protocol.py)

- + corrupt_frame_count: int | -_buffer: bytearray
- + feed(data: bytes): list[TelemetryFrame] | + parse_telemetry_frame(text: str): TelemetryFrame
- + build_command(action: str, target: str, value: str): bytes -> "@CMD:GATE_OPEN;TGT:ENTRYS'r'a"

EdgeDatabase

- path: Path
- backup_directory: Path
- _write_lock: RLock
- + create_transaction(...): EdgeTx
- + bootstrap_chain_head(hash): bool
- + cache_booking(booking)
- + find_active_booking(plate, site)
- + list_due_sync_items(limit)
- + mark_synced(item_id, conf_hash)
- + backup(): Path
- + incomplete_sessions(): list
- + store_incident(payload)

SyncWorker & MqttService

- + cloud_online: bool
- + buffer: deque[tuple]
- + sync_once(client: httpx.Client)
- + publish(suffix, payload, qos)
- + handle_mqtt_command(payload)

PACKAGE apps.web.lib (TypeScript)

<<Service>> Reconciliation

- + computeEdgeIntegrityHash(input)
- + reconcileTransaction(input)
- + runSerializable(operation)
- // Enforces prior hash equality
- // Gross - Tare == Net verification
- // Overload variance calculation
- // Creates Incident on discrepancy

<<Service>> FraudDetection

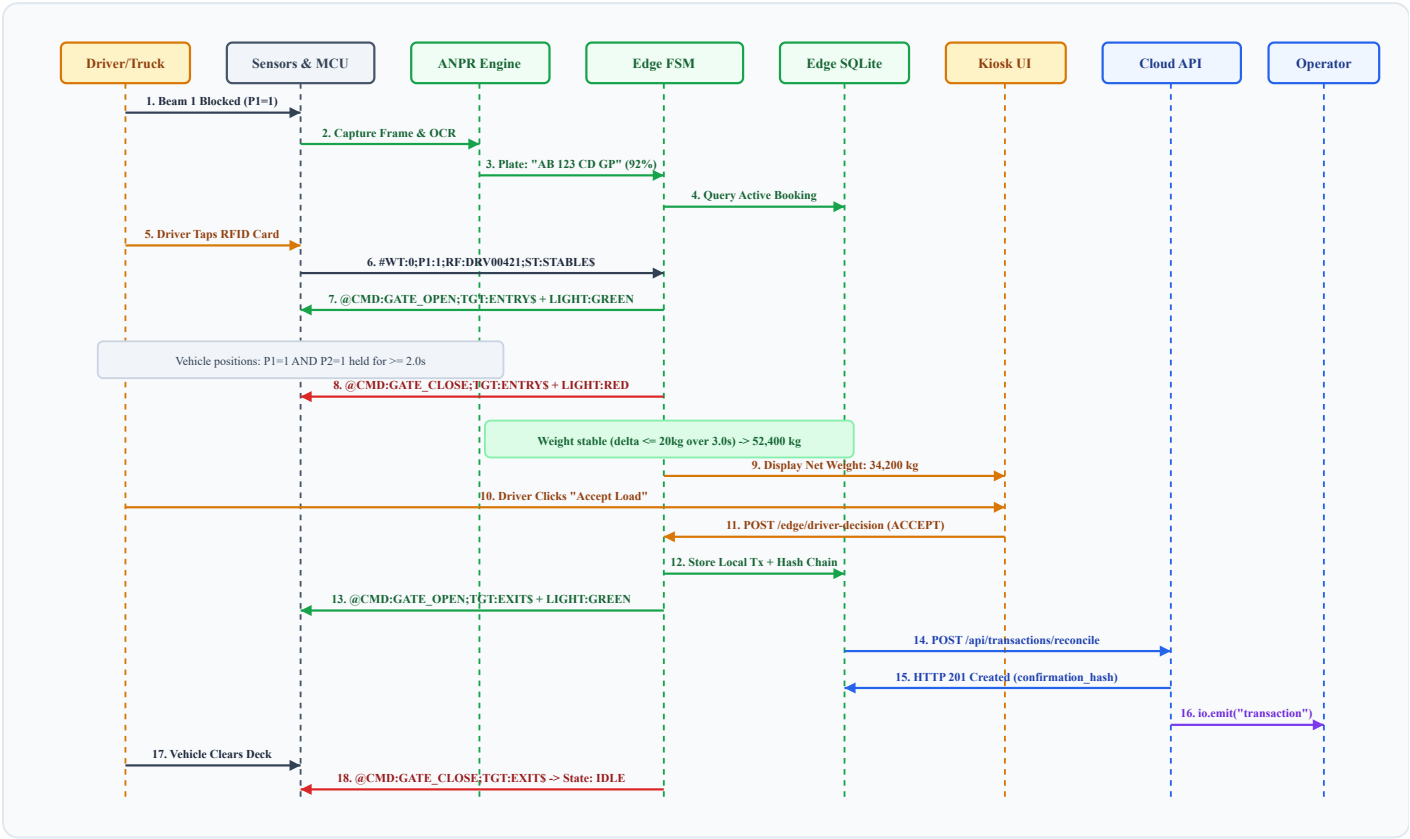
- + runFraudChecks(transactionId)
- checkTareDrift(>500kg)
- checkClonedPlate(travel < 20m)
- checkRouteDeviation(>24h)
- checkCrossSiteVariance(>500kg)
- + incrementAnomalyScore(driver, veh)

<<Security>> Access & RBAC

- + requireRole(roles: UserRole[])
- + requirePermission(permKey: string)
- + requireSiteOrRole(request, roles)
- + isPlatformSuperAdmin(user): bool
- + assertOrganisationActive(org)
- + encryptSensitive / decryptSensitive
- + stableStringify(payload): string

UML Sequence Diagram 1: Automated Normal Weighment Lifecycle

End-to-end event sequence: Vehicle entry, ANPR capture, optical beam hold, load cell capture, kiosk confirmation, and cloud hash reconciliation.



Dual Interlock Verification

Entry gate only opens after matching ANPR plate AND driver RFID tag. Exit gate only opens after stable load capture & driver kiosk sign-off.

Cryptographic Integrity

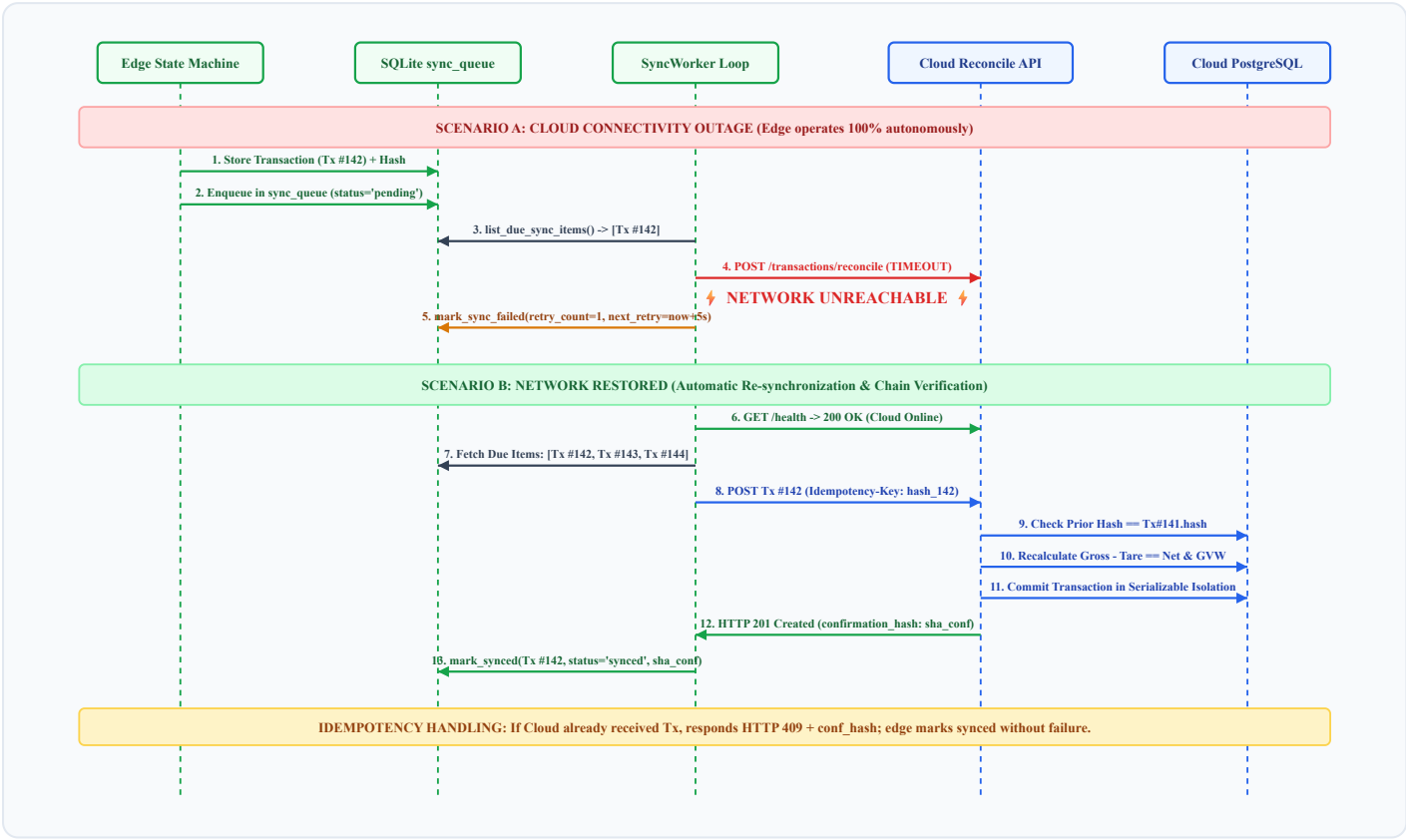
Transaction is signed into the local SHA-256 chain prior to gate release. Cloud verifies prior hash before updating Postgres.

Real-time Telemetry Broadcast

MQTT and Socket.IO pipe 10Hz scale weights and state changes to remote operator dashboards with sub-100ms latency.

UML Sequence Diagram 2: Offline Store-and-Forward Sync & Recovery

72-Hour network outage handling, local SQLite queue buffering, idempotent retry backoff, and hash chain reconciliation.



Exponential Retry Backoff

Sync worker retries failed requests at 5s, 10s, 20s, 40s, and 60s intervals to prevent network flood upon reconnection.

Idempotent Queue Ledger

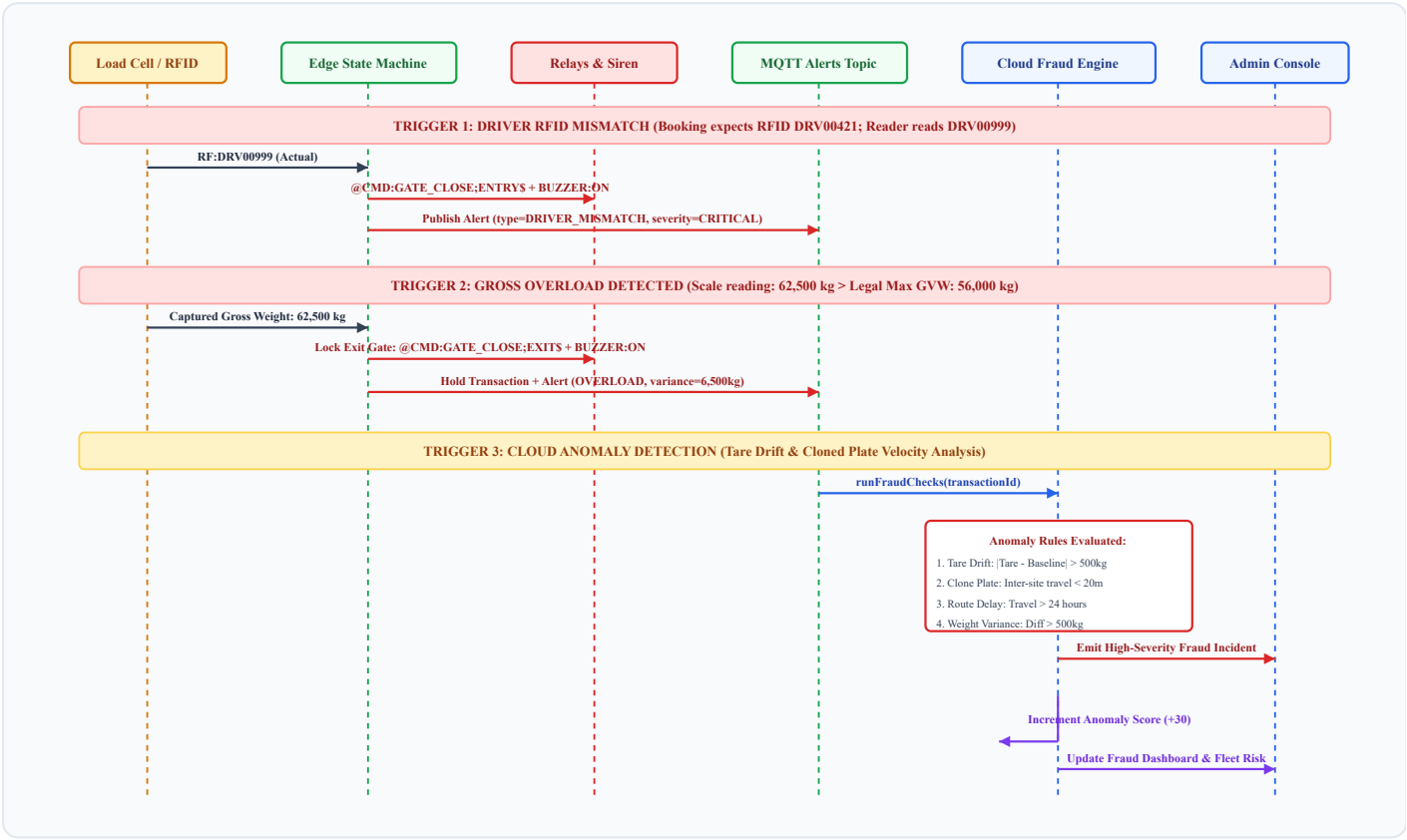
Every payload is tagged with `idempotency_key = integrity_hash`. Retried packets can never produce double transactions in Postgres.

Divergence Alerting

If a 409 error occurs without a confirmation hash (e.g. hash chain broken), a CRITICAL incident is raised for manual auditor inspection.

UML Sequence Diagram 3: Multi-Layer Fraud & Overload Escalation

Real-time fraud triggers: Overload violation, driver RFID mismatch, tare weight drift, and impossible inter-site travel speed.



Physical Safety Interlock

When overload or driver mismatch is detected, the exit boom gate is physically locked and the on-site siren sounds immediately.

Velocity Anomaly Checking

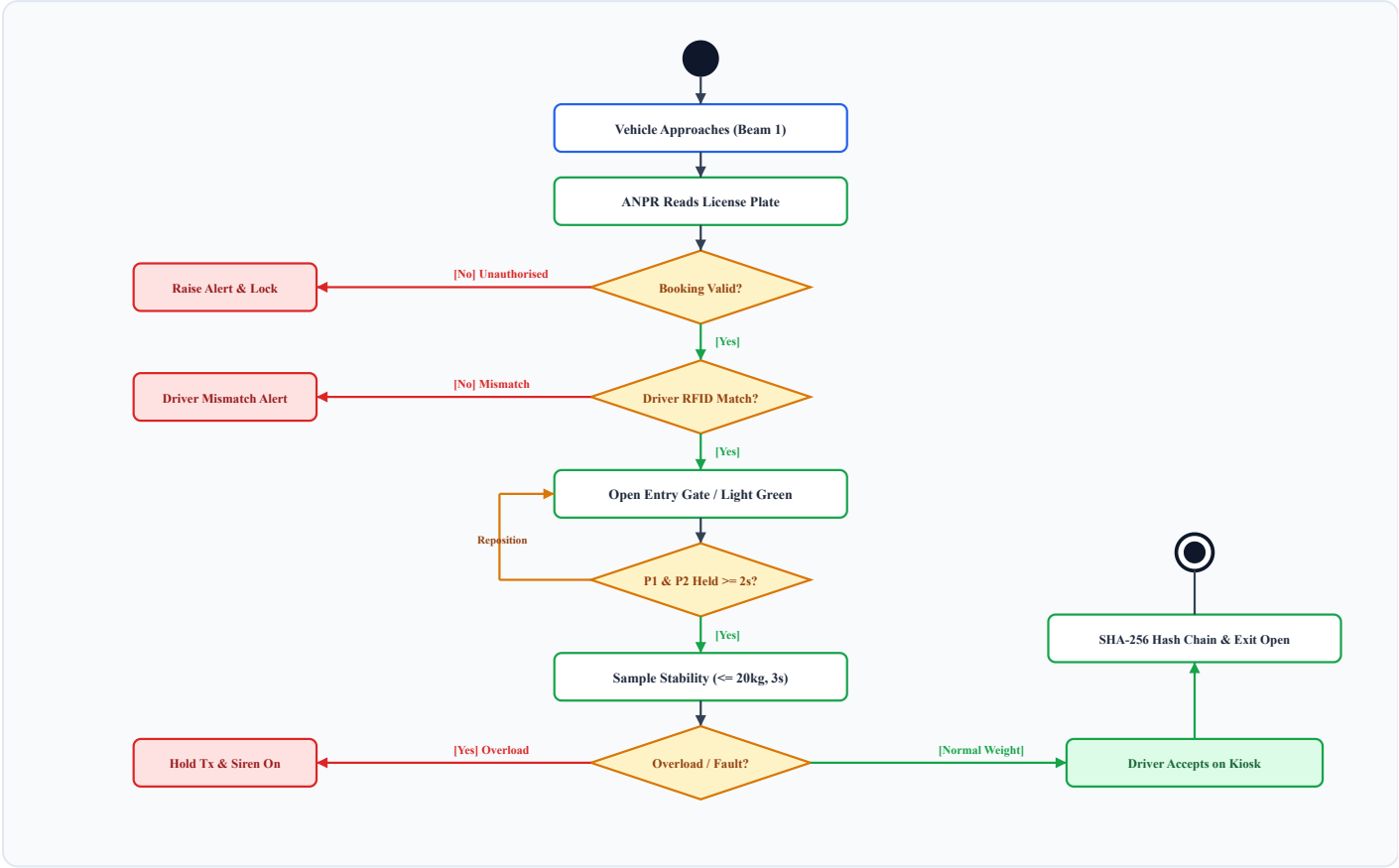
Detects fraudulent cloned plates by computing transit speed between remote mining sites. Travel times < 20 min trigger critical alarms.

Dynamic Risk Profiling

Vehicles and drivers accumulate anomaly scores over time. High-risk assets are automatically flagged for manual physical audits.

UML Activity Diagram: Weighbridge Decision Tree

Deterministic branching logic governing vehicle ingress, sensor validation, stability sampling, overload calculation, and egress.



Automated Normal Flow

Vehicles with valid bookings, verified RFID cards, and stable legal weights complete weighment and exit in < 45 seconds.

Positioning Interlock

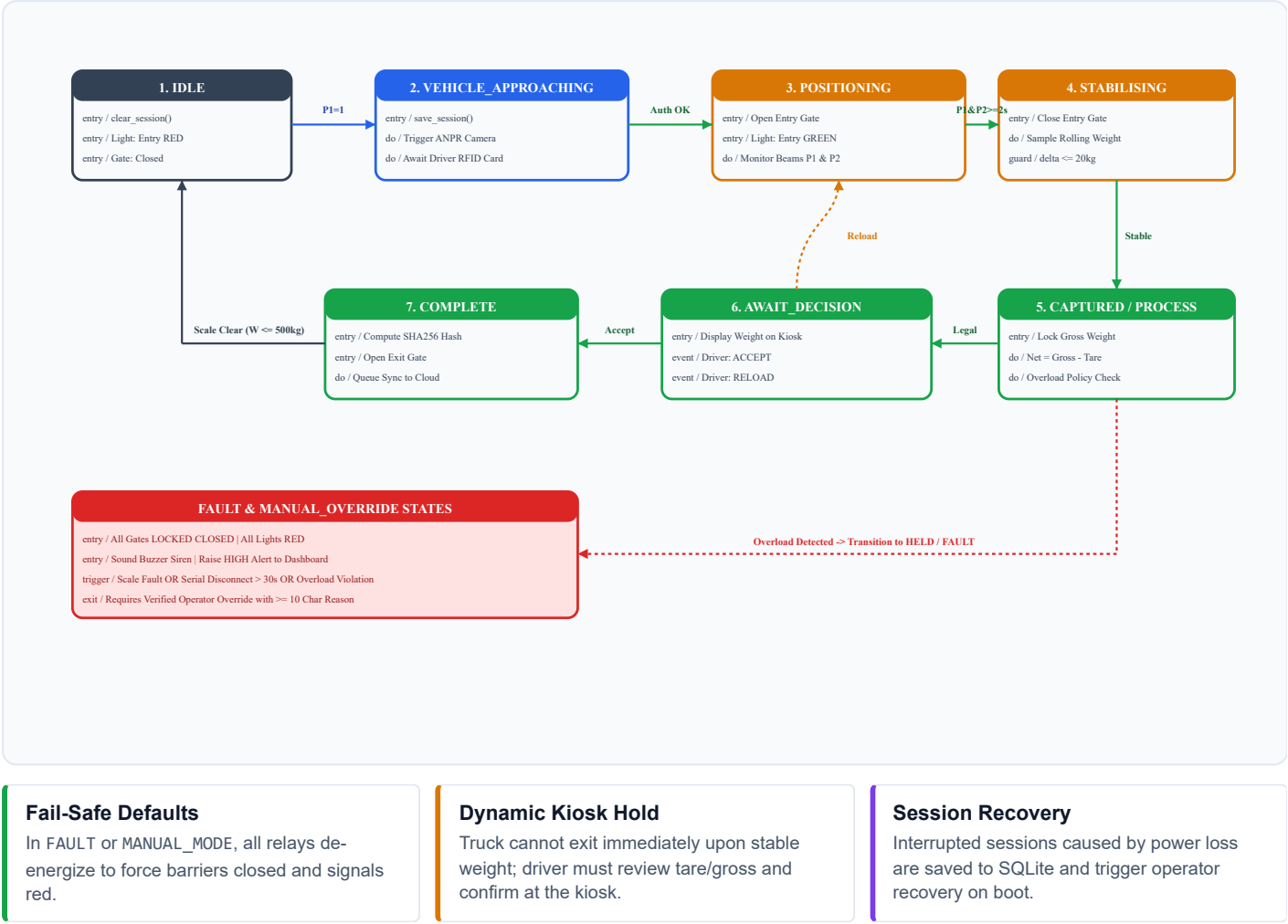
If a driver parks with axles hanging off the scale deck, the state machine refuses to capture and loops back to positioning.

Overload Barrier Locking

Overloaded trucks cannot exit autonomously. The system requires supervisor manual review and offloading before barrier release.

UML State Machine Diagram: 7-Stage Edge Safety Interlocks

Deterministic state transitions, entry actions, exit guards, and fail-safe recovery paths in `apps/site-daemon/state_machine.py`.



Deployment & Network Infrastructure Architecture

Physical fieldbus wiring, opto-isolated industrial relays, on-site edge gateway, and cloud VPC interconnects.

